Panel Data Analysis

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Notations for Panel Data

A double subscript is used to distinguish entities (individuals, states, etc.) and time periods (months, years, etc.):

- i = entity, n = the total number of entities, so $i = 1, \ldots, n$
- t = time period, T = the total number of time periods, so $t = 1, \ldots, T$

Data: Suppose we have k regressors. The data are:

$$(X_{1it}, X_{2it}, \ldots, X_{kit}, Y_{it}), i = 1, \ldots, n, t = 1, \ldots, T .$$

- Balanced panel: no missing observations
- Unbalanced panel: some entities (individuals, states, etc.) are not observed for some time periods (months, years, etc.)

Why Are Panel Data Useful?

With panel data we can control for factors that:

- Are unobserved or unmeasured – and therefore cannot be included in the regression using multiple regression
- Could cause omitted variable bias if they are omitted
- Vary across entities but do not vary over time

Here's the key idea:

If an omitted variable does not change over time, then any changes in Y over time cannot be caused by the omitted variable – the omitted variable can be removed by some methods.

Basic Idea of Using Panel Data

- Panel data lets us eliminate omitted variable bias when the omitted variable is constant over time within a given state
- Intuitively, if an omitted variable is constant over time, then any changes in Y over time cannot be caused by the omitted variable
- Mathematically,if an omitted variable does not change over time, we can remove the omitted variable by taking the first difference of the data across time periods

Panel Data with T = 2

Consider the panel data model

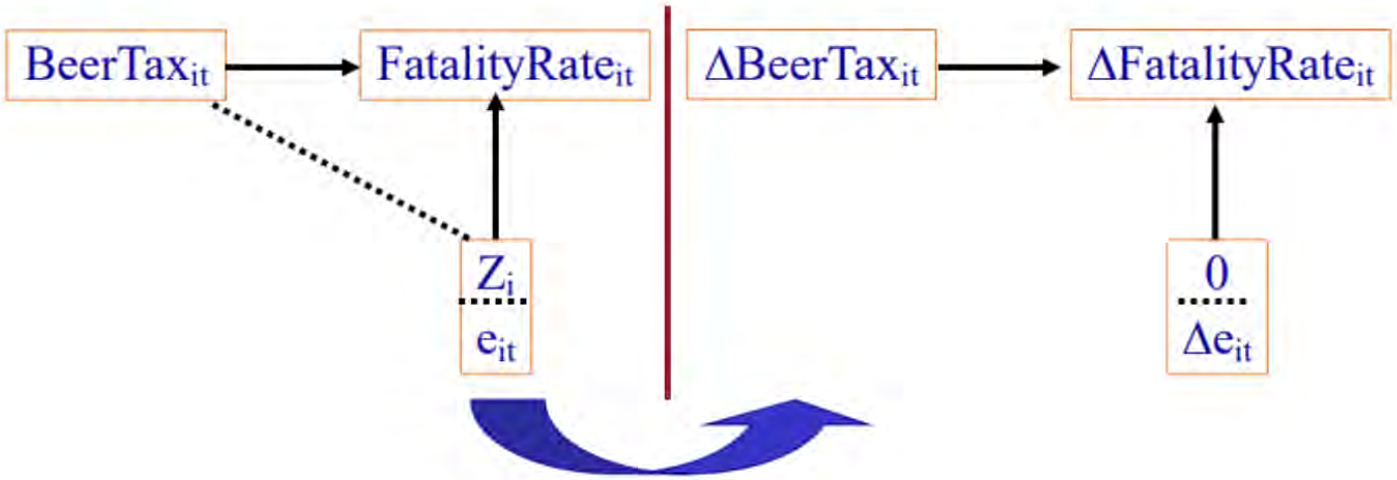
$$\text{FatalityRate}_{it} = \beta_0 + \beta_1 \text{BeerTax}_{it} + \gamma Z_i + e_{it} \ ,$$

where Z_i is a factor that does not change over time, at least during the years on which we have data. (It is reasonable to assume traffic density does not change between two years)

- Suppose Z_i is not observed, so its omission could result in omitted variable bias
- The omitted variable bias can be eliminated using $T = 2$ years

Panel Data with T = 2

The key idea: Any change in the fatality rate from 1982 to 1988 cannot be caused by Z_i, because Z_i (by assumption) does not change between 1982 and 1988.



The omitted variable Z_i doesn't change, so it cannot be a determinant of the change in Y

Panel Data with T = 2

Mathematically,

- When $t = 1982$,

$$\text{FatalityRate}_{i1982} = \beta_0 + \beta_1 \text{BeerTax}_{i1982} + \gamma Z_i + e_{i1982} \,,$$

- When $t = 1988$,

$$\text{FatalityRate}_{i1988} = \beta_0 + \beta_1 \text{BeerTax}_{i1988} + \gamma Z_i + e_{i1988} \,,$$

- Take the difference,

$$\begin{aligned} &\text{FatalityRate}_{i1988} - \text{FatalityRate}_{i1982} \\ &\quad = \beta_1(\text{BeerTax}_{i1988} - \text{BeerTax}_{i1982}) + (e_{i1988} - e_{i1982}) \,. \end{aligned}$$

The omitted variable Z_i has been successfully removed!

Panel Data with T = 2

The difference equation:

$$\begin{aligned}&\text{FatalityRate}_{i1988} - \text{FatalityRate}_{i1982} \\ &\quad = \beta_1(\text{BeerTax}_{i1988} - \text{BeerTax}_{i1982}) + (e_{i1988} - e_{i1982}) .\end{aligned}$$

Suppose $E[e_{it}|\text{BeerTax}_{it}] = 0$. Then

- The new error term, $e_{i1988} - e_{i1982}$, is uncorrelated with either BeerTax_{i1988} or BeerTax_{i1982}
- This "difference" equation can be estimated by OLS, even though Z_i isn't observed. That is, the causal effect β_1 can be estimated by regressing $\text{FatalityRate}_{i1988} - \text{FatalityRate}_{i1982}$ on $\text{BeerTax}_{i1988} - \text{BeerTax}_{i1982}$

Panel Data with T = 2

Mathematically,

- For $t = 1982$,

$$\widehat{\text{FatalityRate}}_{i1982} = \underset{(0.15)}{2.01} + \underset{(0.13)}{0.15} \times \text{BeerTax}_{i1982}, n = 48$$

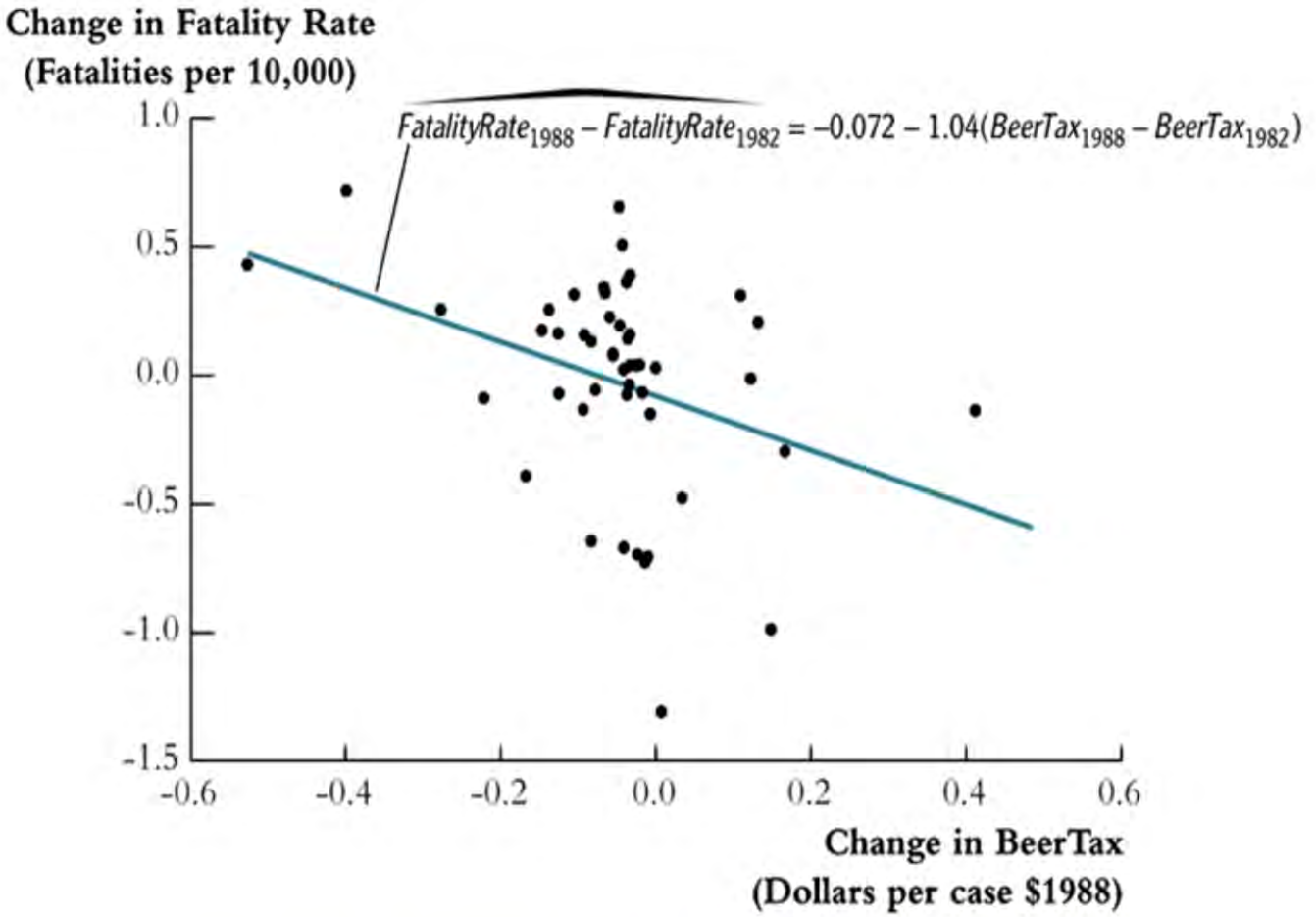
- For $t = 1988$,

$$\widehat{\text{FatalityRate}}_{i1988} = \underset{(0.11)}{1.86} + \underset{(0.13)}{0.44} \times \text{BeerTax}_{i1988}, n = 48$$

- For the difference equation,

$$\widehat{\text{FatalityRate}_{i1988} - \text{FatalityRate}_{i1982}} = \underset{(0.065)}{-0.072} - \underset{(0.36)}{1.04} \times (\text{BeerTax}_{i1988} - \text{BeerTax}_{i1982})$$

Panel Data with T = 2



Higher alcohol taxes, less traffic deaths!

Panel Data with T > 2

What if you have more than 2 time periods ($T > 2$)?

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \gamma Z_i + e_{it}, i = 1, \ldots, n, t = 1, \ldots, T .$$

We can rewrite this in two useful ways:

1. "Fixed effects" regression model
2. "$n - 1$ binary regressor" regression model

Fixed Effects Form

We first rewrite the model in "fixed effects" form. Suppose we have $n = 3$ states: California, Texas, New York.

Population regression for California (that is, $i = CA$):

$$\begin{aligned} Y_{CA,t} &= (\beta_0 + \gamma Z_{CA}) + \beta_1 X_{CA,t} + e_{CA,t} \\ &= \alpha_{CA} + \beta_1 X_{CA,t} + e_{CA,t} \, . \end{aligned}$$

- $\alpha_{CA} = \beta_0 + \gamma Z_{CA}$ doesn't change over time
- α_{CA} is the intercept for CA, and β_1 is the slope
- The intercept is unique to CA, but the slope is the same for all states: parallel lines

Fixed Effects Form

For TX and NY,

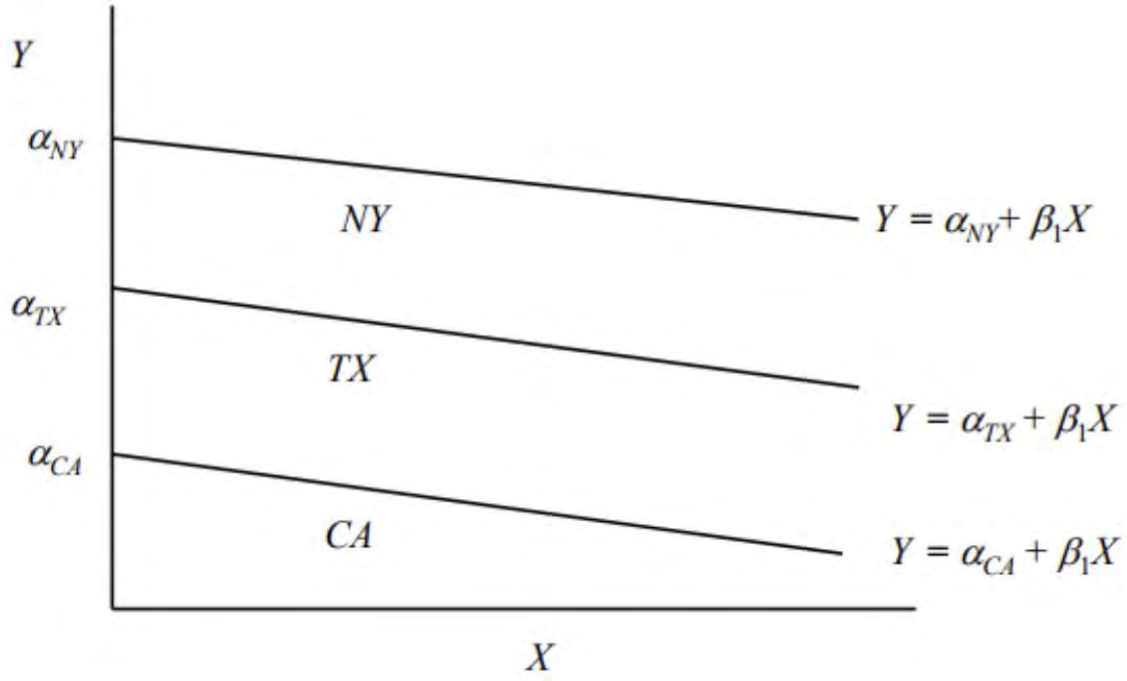
$$Y_{TX,t} = \alpha_{TX} + \beta_1 X_{TX,t} + e_{TX,t}$$
$$Y_{NY,t} = \alpha_{NY} + \beta_1 X_{NY,t} + e_{NY,t} .$$

Collecting the lines for all three states:

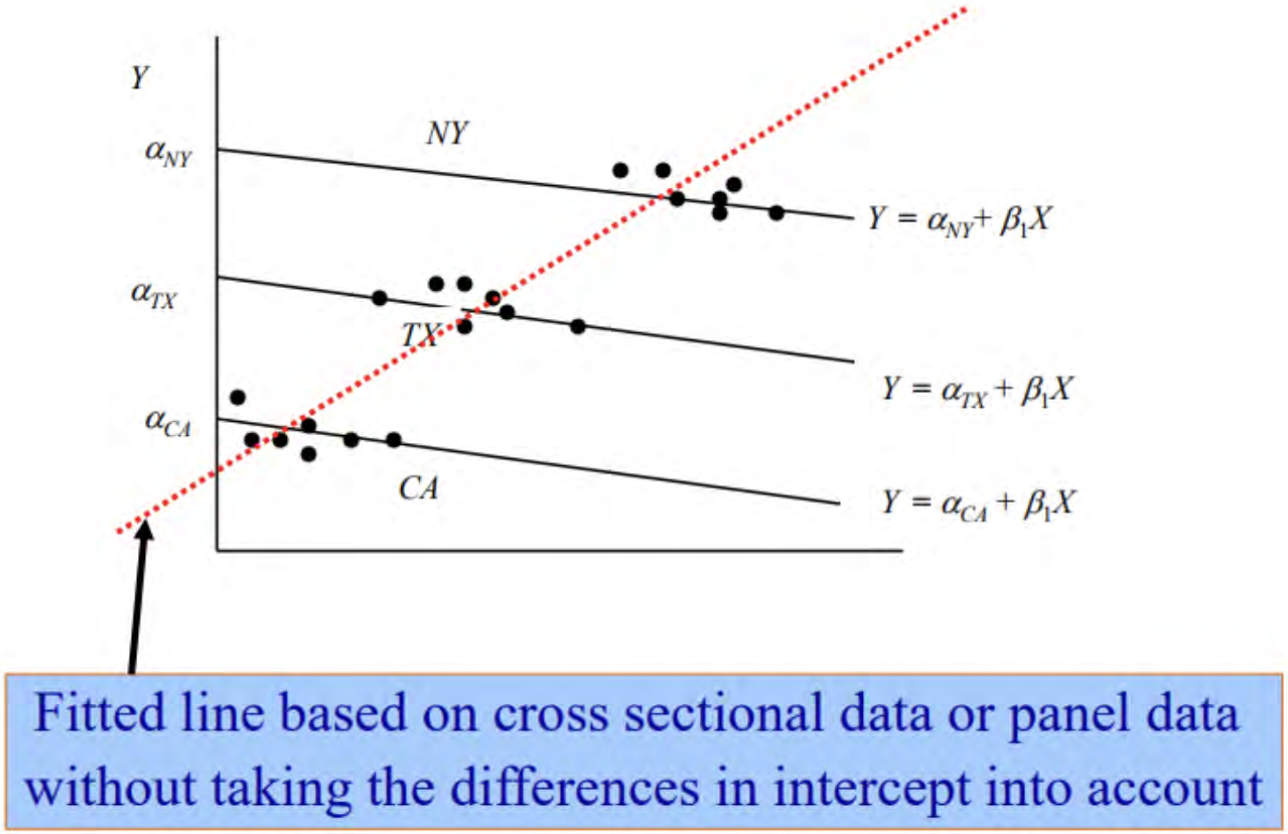
$$Y_{it} = \alpha_i + \beta_1 X_{it} + e_{it} \ , i = CA, TX, NY, t = 1, \ldots, T .$$

- Three parallel regression lines with different intercepts

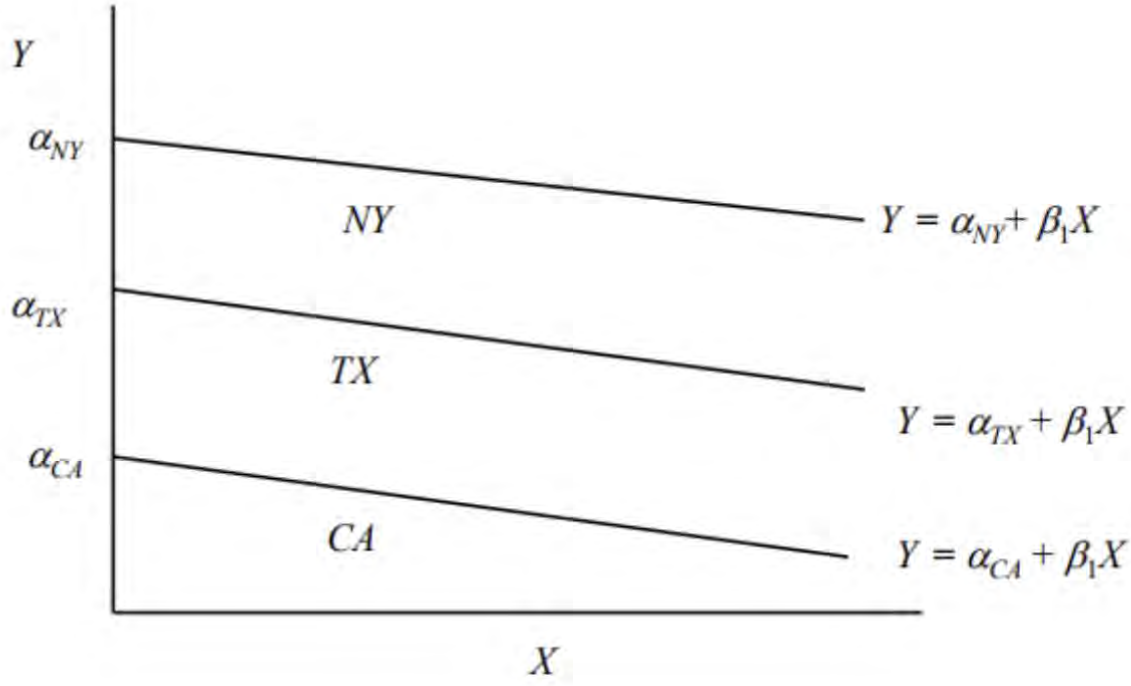
Fixed Effects Form



Fixed Effects Form

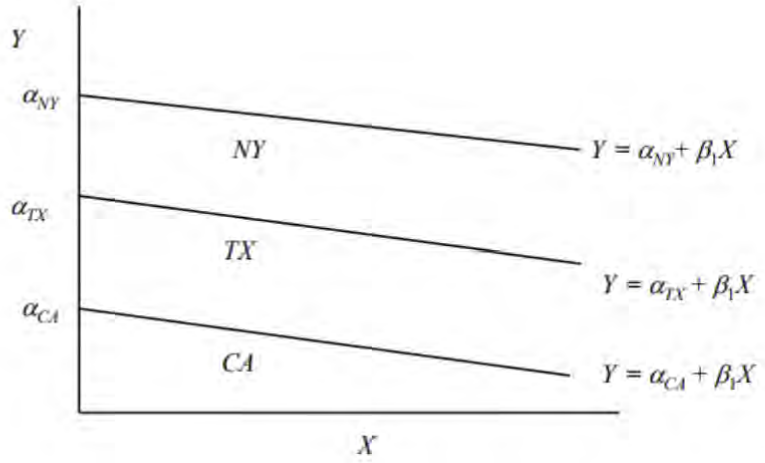


Binary Regressor Form



Recall that shifts in the intercept can be represented using binary regressors (dummy variables)

Binary Regressor Form



In binary (dummy) regressor form:

$$Y_{it} = \beta_0 + \gamma_{CA} DCA_i + \gamma_{TX} DTX_i + \beta_1 X_{it} + e_{it}$$

- $DCA_i = 1$ if state is CA, and 0 otherwise
- $DTX_i = 1$ if state is TX, and 0 otherwise
- Leave out DNY_i (why?)

Binary Regressor Form

$$Y_{it} = \beta_0 + \gamma_{CA} DCA_i + \gamma_{TX} DTX_i + \beta_1 X_{it} + e_{it}$$

	Constant	DCA_i	DTX_i	DNY_i
i=CA:	1	1	0	0
i=TX:	1	0	1	0
i=NY:	1	0	0	1

$$\text{Constant} = DCA_i + DTX_i + DNY_i$$

- Multicollinearity between the constant regressor and the dummy variables
- Solution : choose a base group, say DNY_i and drop it from the regression
- Intercept for NY : β_0; Intercept for CA : $\beta_0 + \gamma_{CA}$; Intercept for TX : $\beta_0 + \gamma_{TX}$

Summary: Two Forms

Fixed effects form:

$$Y_{it} = \beta_1 X_{it} + \alpha_i + e_{it} \ ,$$

where α_i is called a "fixed effect" or "entity fixed effect" or "entity effect" – it is the constant (fixed) effect of being entity i.

Binary regressor form:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \gamma_2 D2_i + \cdots + \gamma_n Dn_i + e_{it} \ ,$$

where $D2_i = 1$ for $i = 2$ (entity 2) and 0 otherwise, etc.

Estimation

Three estimation methods:

1. "$n-1$ binary regressors" OLS regression
2. "Entity-demeaned" OLS regression
3. "Changes" OLS regression

- We already did the "changes" regression (panel data with $T = 2$, 1988 minus 1982)
- Methods #1, #2 and #3 work for general T. (Our textbook may say that method 3 works only for $T = 2$, but this is not quite correct. It actually works for any T)
- Method #1 is only practical when n isn't too big

Binary OLS Regression

Binary regressor form:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \gamma_2 D2_i + \cdots + \gamma_n Dn_i + e_{it} \,. \qquad (1)$$

- First create the binary variables $D2_i, \ldots, Dn_i$
- Then estimate (1) by OLS
- Inference (hypothesis tests, confidence intervals) is as usual (using heteroskedasticity-robust standard errors)
- This is impractical when n is very large (for example if n = 1000 workers)

Demeaned OLS Regression

Fixed effects form:

$$Y_{it} = \beta_1 X_{it} + \alpha_i + e_{it} \ .$$

- Take the entity averages:

$$\frac{1}{T}\sum_{t=1}^{T} Y_{it} = \beta_1 \frac{1}{T}\sum_{t=1}^{T} X_{it} + \alpha_i + \frac{1}{T}\sum_{t=1}^{T} e_{it}$$

- Deviation from the entity averages:

$$Y_{it} - \frac{1}{T}\sum_{t=1}^{T} Y_{it} = \beta_1 (X_{it} - \frac{1}{T}\sum_{t=1}^{T} X_{it}) + e_{it} - \frac{1}{T}\sum_{t=1}^{T} e_{it}$$

[The fixed effects have been successfully removed!]

Demeaned OLS Regression

The demeaned equation:

$$\tilde{Y}_{it} = \beta_1 \tilde{X}_{it} + \tilde{e}_{it}\ , \qquad (2)$$

where $\tilde{Y}_{it} = Y_{it} - \frac{1}{T}\sum_{t=1}^{T} Y_{it}$ is the deviation of Y_{it} from its entity average.

- First construct the demeaned variables $\tilde{Y}_{it}$ and $\tilde{X}_{it}$
- Then estimate (2) by regressing $\tilde{Y}_{it}$ on $\tilde{X}_{it}$
- Inference (hypothesis tests, confidence intervals) is as usual (using heteroskedasticity-robust standard errors)
- This is like the "changes" approach, but instead Y_{it} is deviated from the entity average (rather than the "difference")
- This can be done in a single command in STATA

Use STATA: Empirical Example

```
areg vfrall beertax, absorb(state) r

Linear regression, absorbing indicators                 Number of obs =     336
                                                        F(  1,    287) =   10.41
                                                        Prob > F      =  0.0014
                                                        R-squared     =  0.9050
                                                        Adj R-squared =  0.8891
                                                        Root MSE      =  .18986

------------------------------------------------------------------------------
             |               Robust
      vfrall |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
     beertax |  -.6558736   .2032797    -3.23   0.001    -1.055982   -.2557655
       _cons |   2.377075   .1051515    22.61   0.000     2.170109    2.584041
-------------+----------------------------------------------------------------
       state |   absorbed                                     (48 categories)
```

- "areg" automatically de-means the data
- this is especially useful when n is large
- another useful command "xtreg"

Empirical Example

$$\widehat{\text{FatalityRate}}_{it} = \underset{(0.20)}{-0.66}\text{BeerTax}_{it} + \text{State Effects}, T = 7, n = 48$$

- How many binary regressors would you include to estimate this using the "binary regressor" method?
- Compare slope, standard error to the estimate for the 1988 v.s. 1982 "changes" specification ($T = 2, n = 48$):

$$\widehat{\text{FatalityRate}_{i1988} - \text{FatalityRate}_{i1982}} = \underset{(0.065)}{-0.072} - \underset{(0.36)}{1.04} \times (\text{BeerTax}_{i1988} - \text{BeerTax}_{i1982})$$

Why is there a difference?

Regression with Time Fixed Effects

So far, the panel data can be used to eliminate omitted variable bias due to entity fixed effects, which do not change over time.

An omitted variable might vary over time but not across entities:

- "Safer cars" (air bags, etc.) changes in national laws
- These produce intercepts that change over time
- Let these changes ("safer cars") be denoted by the variable S_t, which changes over time but not across entities
- The resulting population regression model is:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \delta S_t + e_{it}, i = 1, \ldots, n, t = 1, \ldots, T .$$

Regression with Time Fixed Effects

Panel data model with time fixed effects:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \delta S_t + e_{it}, i = 1, \ldots, n, t = 1, \ldots, T \, .$$

We may switch the indexes i and t, and the analysis is similar.

We can rewrite this in two useful ways:

1. "Time fixed effects" regression model:

$$Y_{it} = \beta_1 X_{it} + \mu_t + e_{it} \, ,$$

where μ_t is called a "time fixed effect" or "time effect" – it is the constant (fixed) effect of being in year t.

- The intercepts vary from one period time to next, and the slope is the same for all time periods: parallel lines

Regression with Time Fixed Effects

2. "$T-1$ binary regressor" regression model:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \delta_2 B2_t + \cdots + \delta_T BT_t + e_{it} \ ,$$

where $B2_t = 1$ for $t = 2$ (period 2) and 0 otherwise, etc.

- Include $T-1$ time period binary variables; exclude $B1_t$ to avoid perfect multicolinearity

Three estimation methods:

1. "$T-1$ binary regressors" OLS regression
2. "Time-demeaned" OLS regression
3. "Changes" OLS regression (changes from $i-1$ to i)

Both Entity and Time Fixed Effects

Panel data model with entity and time fixed effects:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \gamma Z_i + \delta S_t + e_{it}, i = 1, \ldots, n, t = 1, \ldots, T .$$

We can rewrite this in two useful ways:

1. "Entity and time effects" regression model:

$$Y_{it} = \beta_1 X_{it} + \alpha_i + \mu_t + e_{it} .$$

2. "$n-1$ and $T-1$ binary regressor" regression model:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \gamma_2 D2_i + \cdots + \gamma_n Dn_i + \delta_2 B2_t + \cdots + \delta_T BT_t + e_{it}$$

Both State and Time Fixed Effects

Estimation:

1. "$n-1$ and $T-1$ binary regressor" OLS regression
   - First create the binary variables $D2_i, \ldots, Dn_i, B2_t, \ldots, BT_t$
   - Then estimate the model by OLS
   - Inference (hypothesis tests, confidence intervals) is as usual (using heteroskedasticity-robust standard errors)
2. "Entity and time-demeaned" OLS regression
   - Deviate Y_{it}, X_{it} from entity averages
   - Deviate the deviations in step 1 from time averages
   - Estimate by OLS using "time- and entity-demeaned" data
   - This can be easily done in STATA
3. "Changes" OLS regression

Use STATA: Empirical Example

```
gen y83=(year==1983)
gen y84=(year==1984)
gen y85=(year==1985)
gen y86=(year==1986)
gen y87=(year==1987)
gen y88=(year==1988)
areg vfrall beertax y83 y84 y85 y86 y87 y88, absorb(state) r

Linear regression, absorbing indicators                Number of obs =     336
                                                       F(  7,   281) =    3.70
                                                       Prob > F      =  0.0008
                                                       R-squared     =  0.9089
                                                       Adj R-squared =  0.8914
                                                       Root MSE      =  .18788
------------------------------------------------------------------------------
             |               Robust
      vfrall |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
     beertax |  -.6399799   .2547149    -2.51   0.013    -1.141371   -.1385884
         y83 |  -.0799029   .0502708    -1.59   0.113    -.1788579    .0190522
         y84 |  -.0724206   .0452466    -1.60   0.111     -.161486    .0166448
         y85 |  -.1239763   .0460017    -2.70   0.007     -.214528   -.0334246
         y86 |  -.0378645   .0486527    -0.78   0.437    -.1336344    .0579055
         y87 |  -.0509021   .0516113    -0.99   0.325    -.1524958    .0506917
         y88 |  -.0518038     .05387    -0.96   0.337    -.1578438    .0542361
       _cons |    2.42847   .1468565    16.54   0.000     2.139392    2.717549
-------------+----------------------------------------------------------------
       state |   absorbed                                     (48 categories)
```

Both State and Time Fixed Effects

- So far, the standard errors are based on the assumption that e_{it} and e_{is} are uncorrelated for $t \neq s$
- If we want to allow for correlation between e_{it} and e_{is} for $t \neq s$, then we have a new formula for computing the standard errors
- STATA does this for us using the cluster option

```
areg vfrall beertax y83 y84 y85 y86 y87 y88, absorb(state) r cluster(state)
```